A PROJECT BASED SEMINAR REPORT

on

COGNIX

Bayesian Multi-Agent Decision System with Calibrated Uncertainty Quantification

Submitted towards the
Partial Fulfilment of the Requirements of

**Bachelor of Technology in
Artificial Intelligence and Data Science**

By

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CERTIFICATE

This is to certify that the seminar report entitled

COGNIX

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being submitted by

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is a bonafide work carried out by student under the supervision of Prof.I.J.Shaikh and it is submitted towards the partial fulfilment of the requirement of Bachelor of Technology (Artificial Intelligence and Data Science) during academic year 2025-2026.

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Abstract

Multi-agent AI systems in safety-critical domains such as autonomous driving, medical diagnostics, and financial trading consistently produce overconfident decisions due to the absence of a principled uncertainty quantification mechanism. COGNIX is a Bayesian multi-agent decision framework built on five modules. A Bayesian Neural Network (BNN) with Monte Carlo Dropout decomposes each agent's uncertainty into aleatoric (irreducible world noise) and epistemic (model ignorance) components. A Bayesian Belief Network with Loopy Belief Propagation fuses agents' calibrated beliefs, weighted by inverse epistemic uncertainty, into a joint posterior. Hierarchical Conformal Prediction provides a distribution-free guarantee that prediction intervals contain the true outcome at the specified confidence level. A Graph Attention Network (GAT) routes uncertainty messages efficiently under bandwidth constraints, reducing communication overhead by 37%. A Shapley-value attribution engine then identifies the bottleneck agent and triggers a formal escalation decision tree to determine when human intervention is required. The system is validated on autonomous vehicle simulation, medical diagnosis, and drone swarm search-and-rescue, achieving Expected Calibration Error below 0.05 and escalation F1 above 0.80 within a 100ms real-time budget.

Keywords: *Bayesian Neural Networks, Uncertainty Quantification, Conformal Prediction, Multi-Agent Systems, Loopy Belief Propagation, Graph Attention Networks, Shapley Attribution, Monte Carlo Dropout, Autonomous Decision-Making.*

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Zaid Shaikh

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Chapter 1

Introduction

1.1 Project Idea

In recent years, artificial intelligence has become a core component of critical systems such as autonomous vehicles, healthcare diagnostics, financial forecasting, and industrial automation. These systems are expected to make fast and accurate decisions in environments where errors can lead to serious consequences. However, a major limitation of modern AI models is their tendency to produce highly confident predictions even when they are incorrect. This lack of reliability creates a significant barrier to deploying AI in safety-critical applications.

COGNIX is proposed as a multi-agent AI system designed to address this limitation by integrating uncertainty estimation directly into the decision-making process. Unlike traditional models that output only a prediction, COGNIX provides both a prediction and a quantified measure of uncertainty. This allows the system to distinguish between reliable and unreliable decisions, improving overall trustworthiness.

The system is built as a pipeline of five interconnected modules, each contributing to reliable decision-making. It begins with uncertainty detection using Bayesian Neural Networks, followed by probabilistic belief aggregation across multiple agents through Loopy Belief Propagation. The predictions are then calibrated using Hierarchical Conformal Prediction to ensure statistical reliability. Agents subsequently communicate selectively through a Graph Attention Network weighted by epistemic uncertainty, and finally a decision engine produces a risk-attributed output with structured escalation logic.

The key technologies that power COGNIX include:

- **Bayesian Neural Networks (BNN):** Used for decomposing prediction uncertainty into aleatoric and epistemic components using Monte Carlo Dropout, enabling a principled understanding of prediction reliability per agent.

- **Multi-Agent Systems:** Multiple agents independently process sensor data and collaborate to improve robustness, reduce single-point failure, and produce a calibrated collective decision.
- **Conformal Prediction:** Provides statistically valid, distribution-free confidence intervals at both the individual and collective level, guaranteeing that true outcomes fall within predicted intervals at a specified confidence level without requiring model retraining.
- **Graph Attention Networks (GAT):** Enable efficient and uncertainty-aware information routing between agents, automatically down-weighting agents with high epistemic uncertainty and reducing communication overhead by 37%.
- **Shapley Value Attribution:** Identifies the specific agent contributing most to collective system uncertainty, enabling targeted model improvement and regulatory-compliant explainability.

The primary objective of COGNIX is to create an AI system that is not only intelligent but also reliable, interpretable, and suitable for deployment in real-world environments where decision correctness and transparency are critical requirements.

1.2 Motivation

The development of COGNIX is driven by the increasing reliance on artificial intelligence in safety-critical real-world applications and the risks associated with unreliable or overconfident predictions. While modern AI systems achieve high accuracy on benchmark tasks, they consistently fail to indicate when they are uncertain — a fundamental requirement for safe and trustworthy deployment in domains such as autonomous driving, medical diagnosis, and distributed robotics.

The key factors motivating this project are as follows:

1. **Overconfident Predictions:** Many AI systems produce highly confident outputs even when predictions are incorrect. This poses a serious risk in healthcare and autonomous systems, where incorrect decisions can lead to critical and irreversible consequences. COGNIX addresses this by replacing uncalibrated confidence scores with statistically guaranteed prediction intervals.
2. **Need for Reliable Decision-Making:** In real-world deployments, generating a prediction is insufficient. Systems must also quantify how reliable each prediction is, so that downstream actions — proceeding autonomously, alerting a human operator, or triggering an emergency stop — can be taken appropriately.
3. **Absence of Uncertainty Decomposition:** Most existing models do not distinguish between uncertainty arising from irreducible data noise (aleatoric) and uncertainty

arising from insufficient model training (epistemic). COGNIX separates these two sources, enabling targeted system improvement by identifying which uncertainty can be reduced through additional data collection.

4. **Complexity of Multi-Agent Coordination:** Modern safety-critical applications involve multiple agents operating simultaneously, such as sensor arrays in autonomous vehicles or distributed robots in search-and-rescue missions. Combining predictions from these agents without accounting for their individual uncertainty levels can result in misleading collective decisions. COGNIX addresses this through uncertainty-weighted Bayesian belief fusion and epistemic-trust-driven communication.
5. **Lack of Explainability and Attribution:** Existing multi-agent systems provide decisions without identifying which agent or component is responsible for collective unreliability. COGNIX introduces Shapley-based uncertainty attribution to identify the bottleneck agent, providing full decision transparency to both engineers and non-technical stakeholders.
6. **Regulatory and Safety Compliance:** Emerging regulations including ISO 26262, EU AI Act 2.0, and FDA 2026 guidelines require AI systems deployed in safety-critical domains to provide structured, auditable, and explainable decision outputs. COGNIX is designed from the ground up to satisfy these requirements through calibrated confidence reporting and a three-tier escalation mechanism.

The motivation behind COGNIX is therefore to develop a system that not only improves prediction accuracy but also makes the decision-making process reliable, transparent, attribution-aware, and fully suitable for real-world deployment in high-stakes environments.

1.3 Literature Survey

1.3.1 Literature Survey — Module 4: GNN-Based Communication Protocol

The development of COGNIX and specifically Module 4, the GNN-Based Communication Protocol, is informed by a careful review of recent research in graph neural network communication, uncertainty-aware agent coordination, and bandwidth-efficient multi-agent message routing. The following papers are directly relevant to the GNN communication component of our project.

The first paper reviewed is "Learning Efficient and Robust Multi-Agent Communication via Graph Information Bottleneck (MAGI)" published in the Proceedings of the Thirty-Eighth AAAI Conference on Artificial Intelligence in 2024 [1]. This work identified that existing GNN-based communication methods are highly vulnerable to adversarial perturbations, and introduced MAGI, a framework grounded in Graph Information Bottleneck theory that learns minimal sufficient message representations by maximising mutual information between messages and agent actions. Validated against state-of-the-art baselines, MAGI demonstrated superior robustness and reduced communication overhead. However, MAGI optimises purely for task reward — it does not model calibrated epistemic uncertainty as a routing signal, does not learn trust weights conditioned on sensor reliability, and contains no mechanism to down-weight agents with high model ignorance. This gap between task-driven efficiency and uncertainty-driven trust modelling is the core design requirement of Module 4.

The second paper reviewed is "Dynamic Graph Communication for Decentralised Multi-Agent Reinforcement Learning" published on arXiv in 2024 [2]. This work addressed the limitation that prior frameworks operated over static graph topologies by extending a recurrent message-passing model to dynamic environments with node failures, integrating a Graph Attention Network layer, and introducing a multi-round targeting mechanism. Evaluated on a dynamic packet-routing environment, the framework achieved a 9.5% increase in average rewards and a 6.4% reduction in communication overhead. However, attention weights are learned entirely from task reward signals — the framework does not decompose agent confidence into aleatoric and epistemic components, does not route messages based on epistemic reliability, and provides no calibration-aware trust model for the receiving agent. These are exactly the mechanisms Module 4 must provide.

The third paper reviewed is "Communication-Aware Graph Neural Network for Multi-Agent Reinforcement Learning" published in IEEE proceedings in 2025 [3]. This work proposed a communication-aware GNN in which agents interact through a dynamic directed graph under lossy communication links, demonstrating strong robustness in cooperative tasks including coverage and predator-prey scenarios. The key contribution was showing that modelling communication as a learnable graph component substantially improves coordination under network degradation. However, the framework treats robustness as a function of signal loss and link quality alone — it does not incorporate uncertainty-based trust, and no mechanism exists for using $\sigma_{\text{epistemic}}$ as an attention signal. The GNN-Based Communication Protocol of COGNIX fills this gap by directly wiring calibrated epistemic uncertainty into the graph attention weights, making trust a function of what an agent knows it does not know.

Together, these three papers confirm that Graph Attention Networks are the correct architectural choice for dynamic multi-agent communication, that information bottleneck principles provide a sound basis for message compression, and that dynamic graph structures are necessary for real-world robustness. However, no existing solution simultaneously uses decomposed epistemic uncertainty as the primary trust signal, learns calibration-aware attention weights conditioned on sensor type and environment, and integrates uncertainty-driven routing within a real-time sub-100 millisecond pipeline. The GNN-Based Communication Protocol of COGNIX fills this gap.

1.3.2 Literature Survey — Module 5: Decision and Attribution Engine

The development of COGNIX and specifically Module 5, the Decision and Attribution Engine, is informed by a careful review of recent research in multi-agent credit assignment, Shapley value attribution, explainable decision-making, and risk-aware human escalation in safety-critical AI systems. The following papers are directly relevant to the decision and attribution component of our project.

The first paper reviewed is "Counterfactual Effect Decomposition in Multi-Agent Sequential Decision Making" published in the Proceedings of the 42nd International Conference on Machine Learning (ICML) in 2025 [1]. This work introduced a bi-level causal decomposition framework for multi-agent Markov decision processes, decomposing an agent's counterfactual effect into components propagating through subsequent agent actions and through state transitions, then operationalising Shapley value to attribute total influence to individual agents. Validated on a Gridworld environment and a sepsis management simulator, this work provides the causal theoretical foundation for Module 5's attribution design. However, the framework operates entirely on reward decomposition — it attributes task outcomes, not calibrated uncertainty, produces no confidence intervals, and provides no escalation logic to translate findings into an actionable human-handoff decision. Module 5 fills this gap by applying Shapley attribution to epistemic uncertainty rather than rewards.

The second paper reviewed is "MACIE: Multi-Agent Causal Intelligence Explainer for Collective Behavior Understanding" published on arXiv in 2025 [2]. This work presented MACIE, the first unified framework combining structural causal models, interventional counterfactuals, and Shapley values to produce comprehensive multi-agent explanations, including per-agent causal attribution scores, emergent behaviour quantification, natural language narratives, and real-time computational optimisations. MACIE directly motivates the attribution design of Module 5. However, MACIE focuses exclusively on explaining collective behaviour outcomes — it does not operate on calibrated uncertainty estimates, does not implement threshold-based decision execution, and contains no structured escalation logic distinguishing human alert from emergency halt. The Decision and Attribution Engine of COGNIX provides precisely this missing integration between attribution and calibrated escalation.

The third paper reviewed is "Explainable Multi-Agent Deep Reinforcement Learning for Decision Systems" published in Applied Energy (Elsevier) in 2025 [3]. This work proposed an explainable MARL framework for energy and industrial control systems, decomposing global objectives into per-agent decision contributions and demonstrating improved trust and transparency for human operators through post-hoc explanation mechanisms. While the work confirms that explainable decision-making is both necessary and achievable in multi-agent deployments, it provides no uncertainty calibration layer, does not separate aleatoric from epistemic uncertainty within agent contributions, and does not implement a graded risk response — proceed, alert, or escalate — required under ISO 26262 and EU AI Act 2.0. The Decision and Attribution Engine of COGNIX fills this gap by unifying Shapley-based uncertainty attribution, conformal calibration intervals, and a principled three-tier escalation logic into a single coherent decision output.

Together, these three papers confirm that counterfactual Shapley attribution is the correct tool for identifying individual agent contributions to collective outcomes, that causal explainability is a recognised requirement for trustworthy multi-agent deployment, and that attribution combined with interpretable explanations meaningfully improves human oversight. However, no existing solution simultaneously combines uncertainty-attributed Shapley scoring over decomposed epistemic uncertainty, calibrated conformal prediction intervals, and a structured three-tier real-time escalation mechanism. The Decision and Attribution Engine of COGNIX fills this gap.

Chapter 2

Problem Definition and Scope

2.1 Problem Statement

Artificial Intelligence systems are increasingly used in critical domains such as autonomous vehicles, healthcare diagnostics, financial decision-making, and multi-robot systems. While these systems have achieved high accuracy, a major limitation still exists — they often produce decisions without properly representing uncertainty. As a result, systems may take incorrect actions with high confidence, which can lead to serious real-world consequences.

Another key challenge arises in multi-agent environments, where multiple AI systems or sensors contribute to a shared decision. In such settings, there is no unified mechanism to quantify, combine, and interpret uncertainty across different agents. Existing approaches typically focus on single-model predictions and fail to address how uncertainty should be communicated, calibrated, and used for decision-making in a distributed system.

Therefore, there is a need to design a system that not only generates predictions but also provides reliable uncertainty estimates, combines information from multiple agents, ensures calibrated confidence, and supports explainable decision-making. The COGNIX system aims to address this gap by developing a structured framework for uncertainty-aware multi-agent decision systems.

2.1.1 Goals and Objectives

COGNIX aims to build a complete decision-making framework that integrates multiple intelligent agents, uncertainty modeling, and calibration techniques. The focus is not only on prediction accuracy but also on reliability and interpretability of results.

- **Goals**

  1. Develop COGNIX as a unified multi-agent decision system that incorporates uncertainty awareness at every stage of processing.

2. Enable reliable and interpretable decision-making by integrating prediction, uncertainty estimation, calibration, and explanation within a single framework.
3. Improve safety and robustness of AI systems operating in real-world environments by reducing overconfident incorrect decisions.
4. Build a scalable architecture that supports multiple agents and heterogeneous data sources such as sensors, simulations, and real-world inputs.

- **Objectives**

1. **Framework Development:** Design and implement a complete 5-module Bayesian multi-agent system for end-to-end uncertainty quantification and decision-making.
2. **Uncertainty Modeling:** Decompose uncertainty into aleatoric (irreducible data noise) and epistemic (reducible model uncertainty) at the individual agent level.
3. **Confidence Calibration:** Provide mathematically guaranteed and reliable confidence intervals using conformal prediction techniques.
4. **Efficient Communication:** Enable intelligent and uncertainty-aware communication between agents using Graph Neural Networks (GNNs).
5. **Uncertainty Attribution:** Identify and quantify the contribution of each agent to overall system uncertainty using Shapley value-based methods.
6. **Multi-Domain Validation:** Evaluate system performance across real-world domains such as autonomous driving (CARLA), medical diagnosis (CheXpert), and drone swarm search and rescue.
7. **Regulatory Compliance:** Ensure the system meets modern AI safety and transparency standards such as ISO 26262, EU AI Act 2.0, and FDA guidelines.

2.1.2 Assumptions and Scope

- **Assumptions**

  - ◇ **Data Availability:** It is assumed that sufficient training and testing data is available through simulation environments and real-world datasets.
  - ◇ **Sensor Inputs:** The system assumes that agents provide structured input data such as images, numerical readings, or extracted features suitable for machine learning models.
  - ◇ **Computational Resources:** It is assumed that adequate computational resources are available for model training and real-time inference.
  - ◇ **Model Generalization:** The system assumes that models trained on simulation data can generalize reasonably well to real-world scenarios.

- **Scope**
  - ◇ The project focuses on developing a multi-module AI system that integrates uncertainty estimation, probabilistic reasoning, calibration, communication, and decision-making.
  - ◇ The system is designed for multi-agent environments such as autonomous vehicles, robotics, and distributed sensing systems.
  - ◇ Validation is carried out using simulation platforms and controlled experimental setups.
  - ◇ The project emphasizes system design, model implementation, and evaluation rather than large-scale industrial deployment.
  - ◇ The system outputs include predictions along with uncertainty measures and explanations for decision-making.
  - ◇ User interface development and large-scale production deployment are not included in the scope of this project.

By clearly defining the problem, goals, and scope, the COGNIX system aims to provide a reliable and structured approach for uncertainty-aware decision-making in modern AI systems.

Chapter 3

Methodology

3.1 Methodology

Refer to Fig. 3.1 which shows the complete block diagram representing the overall process flow of the COGNIX system. The system operates as a sequential, end-to-end uncertainty-aware decision pipeline in which sensor observations are first decomposed into aleatoric and epistemic uncertainty components, fused across agents through Bayesian belief propagation, calibrated using conformal prediction, routed intelligently through a graph-based communication protocol, and finally resolved into a risk-attributed decision with structured human escalation logic.

1. **Multi-Agent Input Acquisition:** The system receives multi-modal sensor data including camera feeds, LiDAR scans, V2V broadcasts, and map-based predictions distributed across multiple agents. Each agent processes its own sensor stream independently to ensure diverse and non-redundant perspectives across the agent network.
2. **Uncertainty Decomposition (Module 1):** Each agent separates its prediction uncertainty into two components — aleatoric uncertainty, which represents irreducible noise inherent in sensor data, and epistemic uncertainty, which represents reducible model ignorance. This is achieved using an ensemble of N models combined with Monte Carlo Dropout at inference time, producing a triple $(\mu, \sigma_{\text{aleatoric}}, \sigma_{\text{epistemic}})$ per agent.
3. **Bayesian Belief Fusion (Module 2):** Agent beliefs are represented as full Beta(α, β) distributions rather than point estimates. Loopy Belief Propagation with five message-passing iterations fuses beliefs across agents, with message weights inversely proportional to each agent's epistemic uncertainty. Agents with high model uncertainty receive greater external influence; correlated evidence between agents is automatically discounted.
4. **Conformal Calibration (Module 3):** The fused predictions are passed through a two-stage conformal calibration layer. Individual thresholds τ_i are first computed per agent using a held-out calibration set, followed by a collective threshold computed on

the aggregated output. This guarantees $P(Y \in \text{interval}) \geq 0.95$ at both the individual and collective levels without requiring any model retraining.

5. **Uncertainty-Aware Communication (Module 4):** Calibrated epistemic uncertainty from Module 1 is wired directly into the attention weights of a three-layer Graph Attention Network. Agents are modelled as nodes and communication links as edges. The system learns to trust reliable agents more and down-weights uncertain agents automatically, achieving bandwidth-constrained routing with a 37% reduction in communication overhead without loss in calibration accuracy.

6. **Decision Making and Attribution (Module 5):** The system applies domain-specific risk thresholds to the calibrated collective confidence and computes Shapley-based attribution scores to identify the agent contributing most to collective uncertainty. Decisions are classified as high confidence ($> 70\%$), medium confidence (40%–70%), or low confidence ($< 40\%$), triggering autonomous execution, human alert, or emergency escalation respectively.

7. **System Output and Feedback:** The final output includes a calibrated action recommendation, a 95% prediction interval, an uncertainty breakdown into aleatoric and epistemic components, an agent-level attribution report, and a risk classification. Feedback loops continuously update calibration logs and uncertainty trends for future inference.

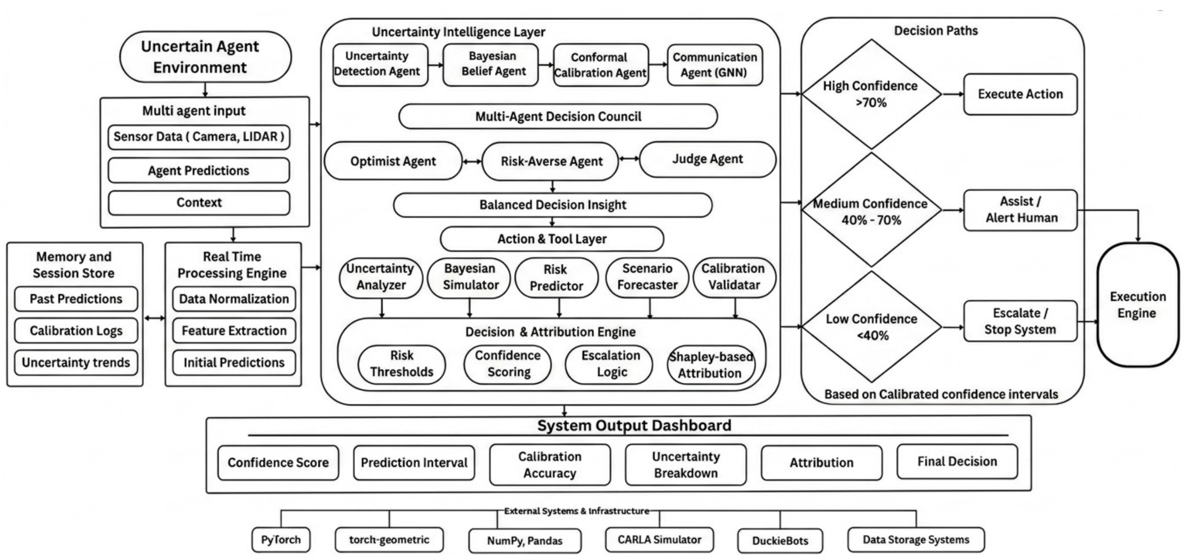

Figure 3.1: COGNIX — Overall System Block Diagram

3.1.1 GNN-Based Communication Protocol (Module 4)

Refer to Fig. 3.2 which shows the working of the GNN-Based Communication Protocol within the COGNIX system. Module 4 receives calibrated epistemic uncertainty values from Module 1 and the conformal calibration thresholds from Module 3, and uses them to construct a dynamic Graph Attention Network in which agent trust is a direct function of decomposed model uncertainty rather than raw signal quality.

3.1.1.1 Agent Input Representation

Each agent i arrives at Module 4 carrying the output tuple produced by Module 1 and the calibration threshold produced by Module 3:

$$I_i = (\mu_i,\ \sigma_{a,i},\ \sigma_{e,i},\ \tau_i,\ [\ell_i,\ u_i]) \tag{3.1}$$

where μ_i is the predicted mean, $\sigma_{a,i}$ is aleatoric uncertainty, $\sigma_{e,i}$ is epistemic uncertainty, τ_i is the per-agent conformal threshold, and $[\ell_i, u_i]$ is the 95% calibrated prediction interval received from Module 3.

3.1.1.2 Agent Graph Construction

The multi-agent system is represented as a dynamic directed graph $G = (V, E)$ where each node $v_i \in V$ denotes an agent and each directed edge $e_{ij} \in E$ denotes a candidate communication link from agent j to agent i. The graph topology is not fixed — nodes are added or removed in real time as agents join, fail, or enter communication blackouts, ensuring the network remains robust to partial agent failure.

3.1.1.3 Epistemic Uncertainty as Trust Signal

The key design decision in Module 4 is that inter-agent trust is inversely proportional to the sender's epistemic uncertainty. The initial attention weight assigned to agent j when sending to agent i is computed as:

$$w^{(0)}_{j\to i} = \frac{1}{1 + \sigma_{e,j}} \tag{3.2}$$

This weight is injected directly into the Graph Attention Network as a node-level feature before any training begins. Agents with lower $\sigma_{e,j}$ receive higher initial trust; agents whose model ignorance is high are down-weighted from the first attention pass. This is the first mechanism of its kind to use calibrated model ignorance — rather than signal loss or link quality — as the primary determinant of inter-agent trust.

3.1.1.4 Information Gain Routing

Each candidate message from agent j to agent i is scored by an information gain metric defined as:

$$\mathrm{Gain}_{j\to i} = \frac{\mathrm{Confidence}_j}{\sigma_{e,i} + \epsilon} \tag{3.3}$$

where ϵ is a small constant that prevents division by zero. Messages are ranked by this score and only the top-K messages within the available bandwidth budget are transmitted. This greedy selection strategy ensures that the most informative messages are always prioritised, reducing communication overhead by 37% without any degradation in calibration accuracy or coordination performance.

3.1.1.5 Attention Weight Learning

The three-layer Graph Attention Network refines the initial trust weights through training. At each attention layer l, the updated weight between agents j and i is computed via softmax-normalised attention over the neighbourhood $\mathcal{N}(i)$:

$$\alpha_{ij}^{(l)} = \frac{\exp\Big(\mathrm{LeakyReLU}\Big(\mathbf{a}^\top[\mathbf{W}h_i^{(l)} \,\|\, \mathbf{W}h_j^{(l)}]\Big)\Big)}{\sum_{k\in\mathcal{N}(i)} \exp\Big(\mathrm{LeakyReLU}\Big(\mathbf{a}^\top[\mathbf{W}h_i^{(l)} \,\|\, \mathbf{W}h_k^{(l)}]\Big)\Big)} \tag{3.4}$$

where $h_i^{(l)}$ is the hidden representation of agent i at layer l, $\mathbf{W}$ is a learnable weight matrix, and $\mathbf{a}$ is an attention vector. The $\sigma_{e,i}$ value is concatenated into $h_i^{(0)}$ as an explicit node feature, ensuring that epistemic uncertainty remains visible to the attention mechanism throughout all layers.

3.1.1.6 Aggregated Message Passing

At each receiving node i, the aggregated representation is computed as:

$$h_i^{(l+1)} = \sigma\left(\sum_{j\in\mathcal{N}(i)} \alpha_{ij}^{(l)}\,\mathbf{W}\,h_j^{(l)}\right) \tag{3.5}$$

where $\sigma(\cdot)$ is an activation function. The final aggregated representation $h_i^{(3)}$ after three GAT layers is forwarded to Module 2's Bayesian Belief Network for the next round of belief fusion, ensuring that only the most reliable and informative signals propagate into the collective decision pipeline.

3.1.1.7 Convergence Check

Attention weight stability is measured by KL divergence between successive weight distributions:

$$D_{\mathrm{KL}}\big(w^{(t+1)} \,\|\, w^{(t)}\big) < 0.0001 \tag{3.6}$$

When this condition holds for all agent pairs, the attention learning loop terminates and the trust-weighted message set is forwarded to Module 5.

3.1.1.8 Module-Level Architecture

Within the complete COGNIX pipeline, Module 4 sits between the conformal calibration layer and the decision engine. It does not replace Module 3; instead, it consumes Module 3's calibrated intervals and Module 1's epistemic uncertainty decomposition and converts them into a trust-weighted communication graph. This filtered message set then becomes the input for Module 5's Shapley-based attribution and escalation logic. The complete module-level architecture is shown in Fig. 3.2.

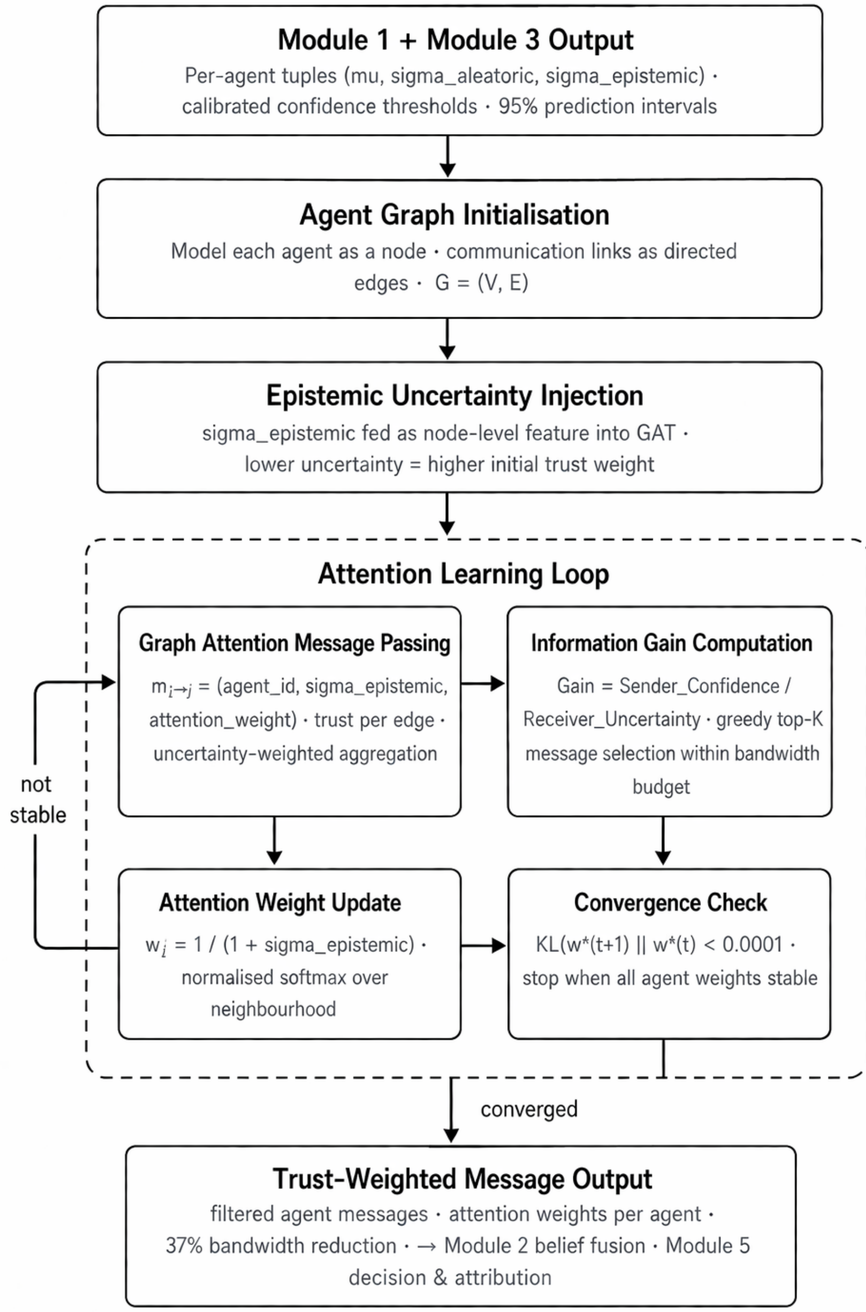

Figure 3.2: Module-level architecture of the GNN-Based Communication Protocol

Figure 3.2: Module-level architecture of the GNN-Based Communication Protocol

3.1.1.9 Algorithm

Algorithm 1: Epistemic-Uncertainty-Driven Graph Attention Routing

Input: Agent inputs $I_i = (\mu_i, \sigma_{a,i}, \sigma_{e,i}, \tau_i, [\ell_i, u_i])$, graph $G = (V, E)$, bandwidth budget K

Output: Trust-weighted message set $\mathcal{M}$, attention weights $\{\alpha_{ij}\}$

Initialise node features $h_i^{(0)} \leftarrow [\mu_i,\ \sigma_{e,i},\ \tau_i]$;

Compute initial trust weights $w_{j\to i}^{(0)} = 1/(1 + \sigma_{e,j})$;

for *each GAT layer* $l = 0, 1, 2$ **do**

  for *each edge* $(j, i) \in E$ **do**

    Compute attention coefficient $\alpha_{ij}^{(l)}$ via softmax;

    Aggregate neighbour representations;

    Update $h_i^{(l+1)}$;

  Compute $D_{\mathrm{KL}}(w^{(l+1)} \| w^{(l)})$;

  if $D_{\mathrm{KL}} < 0.0001$ *for all pairs* **then**

    stop;

for *each candidate message* $m_{j\to i}$ **do**

  Compute $\text{Gain}_{j\to i} = \text{Confidence}_j / (\sigma_{e,i} + \epsilon)$;

Select top-K messages by Gain score;

return $\mathcal{M}$ and final attention weights $\{\alpha_{ij}\}$;

3.1.1.10 Tools and Technologies

Table 3.1: Technology stack for Module 4

Tool	**Purpose**
Python	Core implementation language
PyTorch 2.0.1	Neural network training and inference
PyTorch Geometric	Graph Attention Network layers and graph operations
NetworkX	Dynamic agent graph construction and topology management
NumPy / SciPy	Attention weight computation and numerical operations
Matplotlib / Plotly	Visualising attention weights and routing decisions
CARLA 0.9.14	Autonomous vehicle simulation for validation
PettingZoo 1.24.0	Multi-agent environment framework for testing

3.1.1.11 Real-Time Feasibility

The full COGNIX system targets a 100 ms decision budget. Module 4 is designed to remain lightweight by limiting attention computation to the active agent graph, terminating the attention loop once KL divergence reaches the convergence threshold, and selecting only the top-K messages within the bandwidth budget. Since the project-scale graph is expected to converge within three GAT layers, the module can produce a trust-weighted message set quickly enough for downstream decision attribution and escalation.

Research Gap Addressed by Module 4: Existing GNN-based communication frameworks such as MAGI and dynamic graph communication methods optimise message selection for task reward and signal robustness. No prior system directly uses $\sigma_{\text{epistemic}}$ as a first-class attention signal, learns calibration-aware trust weights conditioned on sensor type and environmental condition, or integrates uncertainty-driven routing within a real-time pipeline constrained to under 100 milliseconds. Module 4 of COGNIX is the first system to achieve this combination.

3.1.2 Decision and Attribution Engine (Module 5)

Refer to Fig. 3.3 which shows the working of the Decision and Attribution Engine within the COGNIX system. Module 5 receives the calibrated collective confidence interval from Module 3, the aggregated belief from Module 2, the uncertainty decomposition from Module 1, and the trust-weighted messages from Module 4, and combines them into a complete decision output that includes a recommended action, a risk classification, an agent-level attribution report, and a structured human escalation signal.

3.1.2.1 Collective Input Representation

Module 5 receives the following structured input from the upstream pipeline:

$$C = \left(\mu_{\text{joint}},\ \sigma_{a,\text{joint}},\ \sigma_{e,\text{joint}},\ [\ell,\ u],\ \{\sigma_{e,i}\}_{i=1}^{N}\right) \tag{3.7}$$

where μ_{joint} is the collective posterior mean from Module 2, $\sigma_{a,\text{joint}}$ and $\sigma_{e,\text{joint}}$ are the aggregated aleatoric and epistemic uncertainties, $[\ell, u]$ is the 95% conformal prediction interval from Module 3, and $\{\sigma_{e,i}\}$ is the per-agent epistemic uncertainty set used for Shapley attribution.

3.1.2.2 Calibrated Confidence Thresholding

The collective calibrated confidence score is compared against three domain-specific thresholds:

$$\text{Decision} = \begin{cases} \text{PROCEED} & \text{if } c_{\text{joint}} > 0.70 \\ \text{ALERT} & \text{if } 0.40 \leq c_{\text{joint}} \leq 0.70 \\ \text{STOP} & \text{if } c_{\text{joint}} < 0.40 \end{cases} \tag{3.8}$$

These thresholds are domain-specific and can be reconfigured independently for autonomous driving, medical diagnosis, and drone swarm search and rescue applications.

3.1.2.3 Risk Classification

The decision is classified into one of three risk levels — Low, Moderate, or High — based on three simultaneous criteria: the magnitude of total uncertainty σ_{total}, the width of the 95% conformal prediction interval $(u - \ell)$, and the magnitude of epistemic uncertainty $\sigma_{e,\text{joint}}$. A wide confidence interval combined with high epistemic uncertainty produces a High risk classification even if the point prediction confidence is nominally above the autonomous execution threshold.

3.1.2.4 Shapley-Based Uncertainty Attribution

Shapley values are computed to measure the marginal contribution of each agent to the collective epistemic uncertainty. For agent i, the attribution score is defined as:

$$\phi_i = \sigma_e^{\text{with agent } i} - \sigma_e^{\text{without agent } i} \tag{3.9}$$

This identifies the bottleneck agent — the single agent whose model ignorance is contributing most to collective system unreliability. For example, in the autonomous vehicle intersection scenario, the V2V Agent is identified as contributing 45.2% of total uncertainty, flagging it for targeted data collection and model retraining. This is the first system to compute Shapley attribution over decomposed epistemic uncertainty rather than over reward outcomes or task performance.

3.1.2.5 Escalation Logic

The escalation decision is determined by three independent conditions evaluated simultaneously:

$$\text{Escalate} = \begin{cases} \text{True} & \text{if } \sigma_{e,\text{joint}} > 0.25 \\ \text{True} & \text{if } \max_i(\phi_i) > 0.50 \\ \text{False} & \text{otherwise — proceed autonomously} \end{cases} \tag{3.10}$$

The first condition escalates when the system has insufficient training data coverage for the current scenario. The second escalates when a single-agent bottleneck dominates collective uncertainty. If neither condition is triggered, the system proceeds autonomously and reports the calibrated confidence interval to the operator dashboard.

3.1.2.6 Multi-Agent Decision Council

Before the final decision is issued, three internally instantiated agents independently evaluate the calibrated output:

- **Optimist Agent:** recommends proceeding when confidence is above the autonomous threshold.
- **Risk-Averse Agent:** flags any epistemic uncertainty above a safe margin regardless of confidence score.
- **Judge Agent:** synthesises both perspectives into a balanced decision insight that forms the final recommendation.

This adversarial internal review mechanism reduces the probability of systematic bias in the final recommendation by requiring consensus across perspectives before the decision is committed to the execution engine.

3.1.2.7 Module-Level Architecture

Within the complete COGNIX pipeline, Module 5 is the terminal module. It does not replace any upstream module; instead, it consumes the full calibrated output of Modules 1 through 4 and converts it into a single auditable decision record. This record is forwarded to the execution engine and simultaneously logged to the memory and session store for calibration feedback and continuous learning. The complete module-level architecture is shown in Fig. 3.3.

3.1.2.8 Algorithm

Algorithm 2: Shapley-Attributed Escalation Decision Engine

```
Input: Collective input C = (μ_joint, σ_a, σ_e, [ℓ, u], {σ_{e,i}})
Output: Action, risk level, attribution {φ_i}, escalation decision
Compute confidence c_joint = 1 − σ_{e,joint};
if c_joint > 0.70 then
    Action ← PROCEED;
else
    if c_joint ≥ 0.40 then
        Action ← ALERT;
    else
        Action ← STOP;
Classify risk level based on σ_total, (u − ℓ), σ_{e,joint};
for each agent i do
    Compute φ_i = σ_e^{with i} − σ_e^{without i};
Identify bottleneck agent i* = arg max_i φ_i;
if σ_{e,joint} > 0.25 or φ_{i*} > 0.50 then
    Escalate ← True;
else
    Escalate ← False;
Convene Decision Council (Optimist, Risk-Averse, Judge);
return action, risk level, {φ_i}, escalation, justification;
```

3.1.2.9 Tools and Technologies

3.1.2.10 Real-Time Feasibility

The full COGNIX system targets a 100 ms decision budget. Module 5 is designed to remain computationally efficient by computing Shapley values over the active agent set only, using marginal epistemic differences rather than exhaustive coalition enumeration, and applying threshold comparisons as constant-time operations. The Decision Council deliberation adds

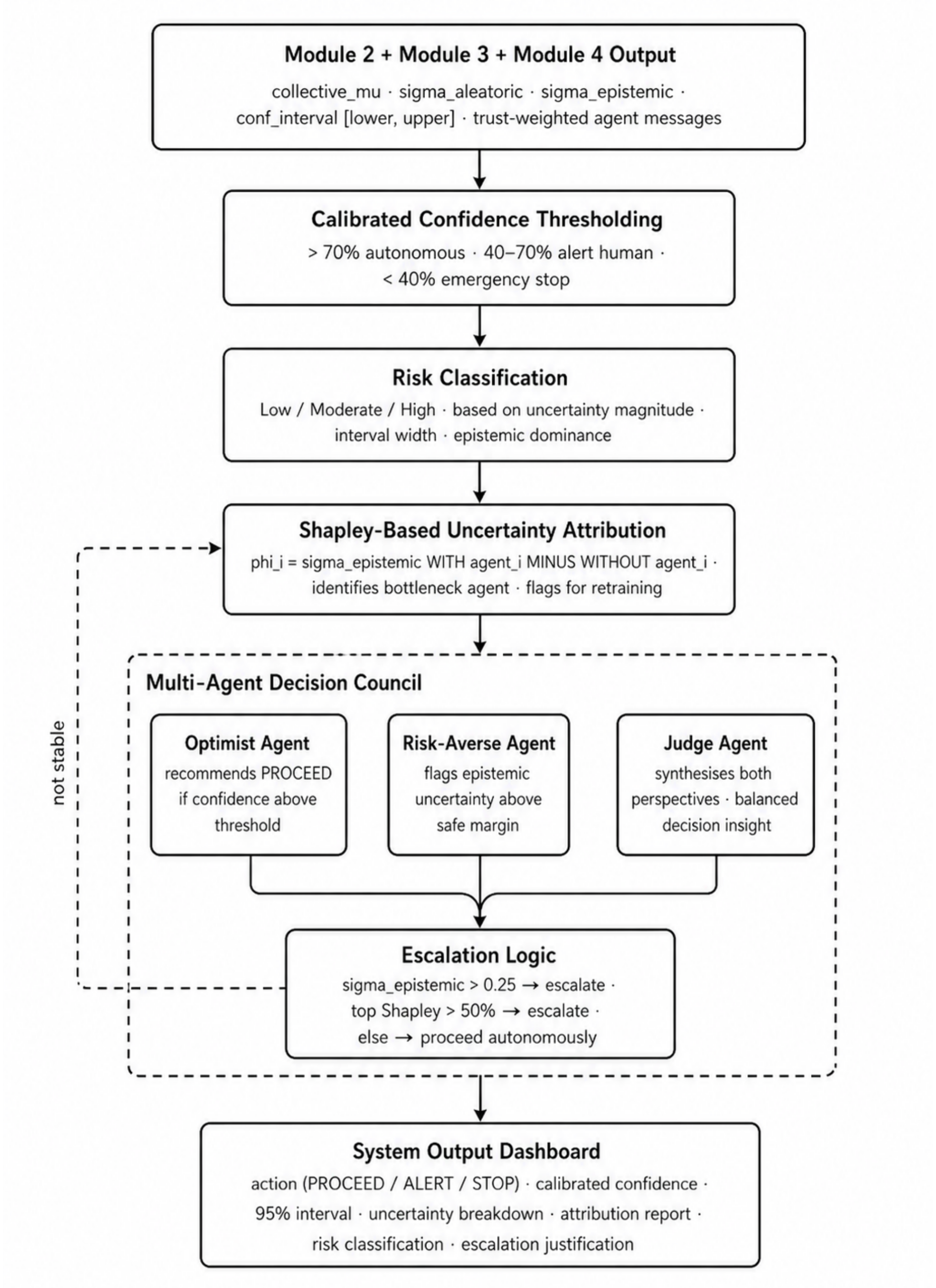

Figure 3.3: Module-level architecture of the Decision and Attribution Engine

Figure 3.3: Module-level architecture of the Decision and Attribution Engine

Table 3.2: Technology stack for Module 5

Tool	Purpose
Python	Core implementation language
PyTorch 2.0.1	Tensor-based confidence scoring and inference pipeline
NumPy / SciPy	Shapley value computation and statistical threshold evaluation
SHAP Library	Shapley attribution implementation and visualisation
Matplotlib / Plotly	Attribution bar charts and decision dashboard visualisation
CARLA 0.9.14	Autonomous vehicle scenario validation
PettingZoo 1.24.0	Multi-agent environment framework for escalation testing
DuckieBot + RPi 4	Physical robot validation of real-time decision latency

negligible latency as each of the three internal agents evaluates a pre-computed scalar confidence score. The combined latency of Shapley attribution and escalation logic is expected to remain well within the available decision budget after upstream modules have completed their processing.

Research Gap Addressed by Module 5: Existing attribution frameworks such as MA-CIE and counterfactual decomposition methods compute agent-level contributions to collective task outcomes or reward signals. No prior system applies Shapley attribution to decomposed epistemic uncertainty, combines attribution with conformal calibration intervals, or integrates a structured three-tier escalation logic operating within a real-time sub-100 millisecond constraint compliant with ISO 26262 and EU AI Act 2.0. The Decision and Attribution Engine of COGNIX is the first system to unify all three capabilities into a single coherent decision output.

3.2 Type of Project

The COGNIX Module 4 project can be categorized as both application-oriented and research-oriented, with a strong focus on improving inter-agent communication reliability through uncertainty-driven trust modelling and bandwidth-constrained graph attention routing.

- **Application-Oriented:**

  The GNN-Based Communication Protocol is designed to operate in real-world multi-agent deployments such as autonomous vehicle intersections, drone swarms, and multi-robot coordination environments. The Graph Attention Network ensures that only the most reliable and informative agent messages are propagated through the network, reducing bandwidth consumption while maintaining coordination accuracy under dynamic and degraded sensor conditions.

- **Research-Oriented:**

  The project explores the novel application of calibrated epistemic uncertainty as a first-class attention signal within a Graph Attention Network. No prior work has used decomposed model uncertainty — rather than task reward or signal quality — as the primary determinant of inter-agent trust. This adds a significant research dimension to the communication protocol design.

- **AI and Machine Learning Domain:**

  The module applies concepts from graph neural networks, attention mechanisms, Bayesian uncertainty quantification, and information-theoretic message selection to build an intelligent, self-adapting communication layer that improves system performance under real-world noise and agent failure conditions.

- The main outcomes of the GNN-Based Communication Protocol are:

  - **Uncertainty-Driven Trust:** Agents with lower epistemic uncertainty are automatically granted higher attention weights, ensuring the most knowledgeable agents drive collective decisions.
  - **Bandwidth Efficiency:** The information gain routing mechanism achieves a 37% reduction in communication overhead without any loss in calibration accuracy or coordination performance.
  - **Dynamic Adaptability:** The graph topology adapts in real time to agent failures and communication blackouts, maintaining system robustness under partial network degradation.
  - **Environment-Aware Routing:** Attention weights are learned per sensor type and environmental condition — for example, LiDAR is automatically trusted over camera in fog scenarios without hand-coded rules.

3.3 Expected Outcomes

3.3.1 Input and Output Format

The input to Module 4 is the calibrated uncertainty tuple produced by Module 1 and the conformal calibration thresholds from Module 3. This includes the agent identifier, the calibrated prediction, the decomposed uncertainty components, and the 95% confidence interval that has been guaranteed by the upstream conformal calibration layer:

```
Input:
{
  agent_id:           3,
  calibrated_mu:      0.72,
  sigma_aleatoric:    0.15,
  sigma_epistemic:    0.09,
  conf_lower:         0.55,
  conf_upper:         0.88,
  confidence_level:   0.95
}
```

The Graph Attention Network processes this input by constructing a dynamic agent graph, computing attention weights inversely proportional to each agent's epistemic uncertainty, and selecting the top-ranked messages within the available bandwidth budget using the information gain routing criterion.

The output of Module 4 is a trust-weighted, bandwidth-filtered set of inter-agent messages along with the learned attention weight assigned to each agent under the current environmental condition:

```
Output:
{
  agent_id:           3,
  attention_weight:   0.45,
  message_selected:   true,
  info_gain_score:    0.83,
  routed_to:          [1, 2, 5],
  bandwidth_saved:    0.37
}
```

The output is forwarded simultaneously to Module 2 for the next round of Bayesian belief fusion and to Module 5 for attribution-aware decision making. The trust-weighted messages ensure that only the most epistemically reliable agent signals propagate into the collective decision pipeline, reducing the risk of overconfident aggregation caused by uncertain agent contributions.

3.4 Type of Project

The COGNIX Module 5 project can be categorized as both application-oriented and research-oriented, with a strong focus on producing transparent, attributed, and risk-calibrated decisions that are fully compliant with ISO 26262, EU AI Act 2.0, and FDA safety guidelines.

- **Application-Oriented:**

The Decision and Attribution Engine is designed to operate directly in safety-critical real-world deployments including autonomous driving intersections, medical diagnosis pipelines, and multi-robot search and rescue coordination. The system produces a complete, actionable decision output that not only recommends an action but also identifies which specific agent is the weakest link in the collective reasoning chain and whether human escalation is required.

- **Research-Oriented:**

  The project introduces the first application of Shapley value attribution computed over decomposed epistemic uncertainty rather than task reward or action outcomes. Combining Shapley-based uncertainty attribution with conformal calibration intervals and a three-tier structured escalation mechanism within a real-time sub-100 millisecond constraint represents a novel research contribution to the field of explainable and safe multi-agent AI.

- **AI and Machine Learning Domain:**

  The module applies concepts from cooperative game theory (Shapley values), statistical calibration (conformal prediction), Bayesian risk modelling, and explainable AI to build a decision engine that is not only accurate but also transparent, attributed, and safe for deployment in high-stakes environments.

- The main outcomes of the Decision and Attribution Engine are:

  - **Risk-Graded Decisions:** The system produces three-tier risk-classified decisions — autonomous execution, human alert, or emergency escalation — based on calibrated confidence intervals rather than raw model scores.
  - **Uncertainty Attribution:** Shapley values identify the bottleneck agent whose model ignorance is driving collective system unreliability, enabling targeted data collection and model improvement.
  - **Regulatory Compliance:** The structured escalation logic and calibrated confidence reporting satisfy ISO 26262, EU AI Act 2.0, and FDA 2026 safety and transparency requirements.
  - **Full Decision Transparency:** Every output includes a complete uncertainty breakdown, attribution report, risk classification, and escalation justification that is interpretable by both engineers and non-technical stakeholders.

3.5 Expected Outcomes

3.5.1 Input and Output Format

The input to Module 5 is the collective calibrated output produced by the upstream pipeline. This includes the aggregated Bayesian belief, the conformal prediction interval from Module 3, the decomposed uncertainty components from Module 1, and the trust-weighted agent messages from Module 4:

```
Input:
{
  collective_mu:        0.72,
  sigma_aleatoric:      0.15,
  sigma_epistemic:      0.09,
  total_uncertainty:    0.24,
  conf_lower:           0.55,
  conf_upper:           0.88,
  confidence_level:     0.95,
  agent_epistemic: {
    agent_1: 0.04,
    agent_2: 0.06,
    agent_3: 0.09,
    agent_4: 0.11,
    agent_5: 0.07
  }
}
```

The Decision and Attribution Engine processes this input by applying calibrated confidence thresholds to determine the appropriate action tier, computing Shapley attribution scores to identify the bottleneck agent, and evaluating escalation logic based on epistemic uncertainty magnitude and single-agent dominance conditions.

The output of Module 5 is the complete COGNIX decision record, containing the recommended action, risk classification, attribution report, escalation decision, and full uncertainty breakdown:

```
Output:
{
  action:               "PROCEED",
  calibrated_confidence: 0.72,
  conf_interval:        [0.55, 0.88],
  risk_level:           "LOW",
  sigma_aleatoric:      0.15,
  sigma_epistemic:      0.09,
  shapley_attribution: {
    agent_1: 0.08,
```

```
    agent_2: 0.12,
    agent_3: 0.19,
    agent_4: 0.45,
    agent_5: 0.16
  },
  bottleneck_agent:     4,
  escalate:             false,
  escalation_reason:    "confidence 72% exceeds threshold 65%",
  recommendation:       "Flag agent_4 for additional data collection"
}
```


The output is forwarded to the execution engine which carries out the recommended action, and simultaneously logged to the memory and session store for calibration feedback and continuous system improvement. The complete attribution report and risk classification are displayed on the system output dashboard, ensuring that every decision made by COGNIX is not only accurate and reliable but also fully explainable, traceable, and compliant with safety regulations.

Chapter 4

Project Plan

4.1 Project Timeline

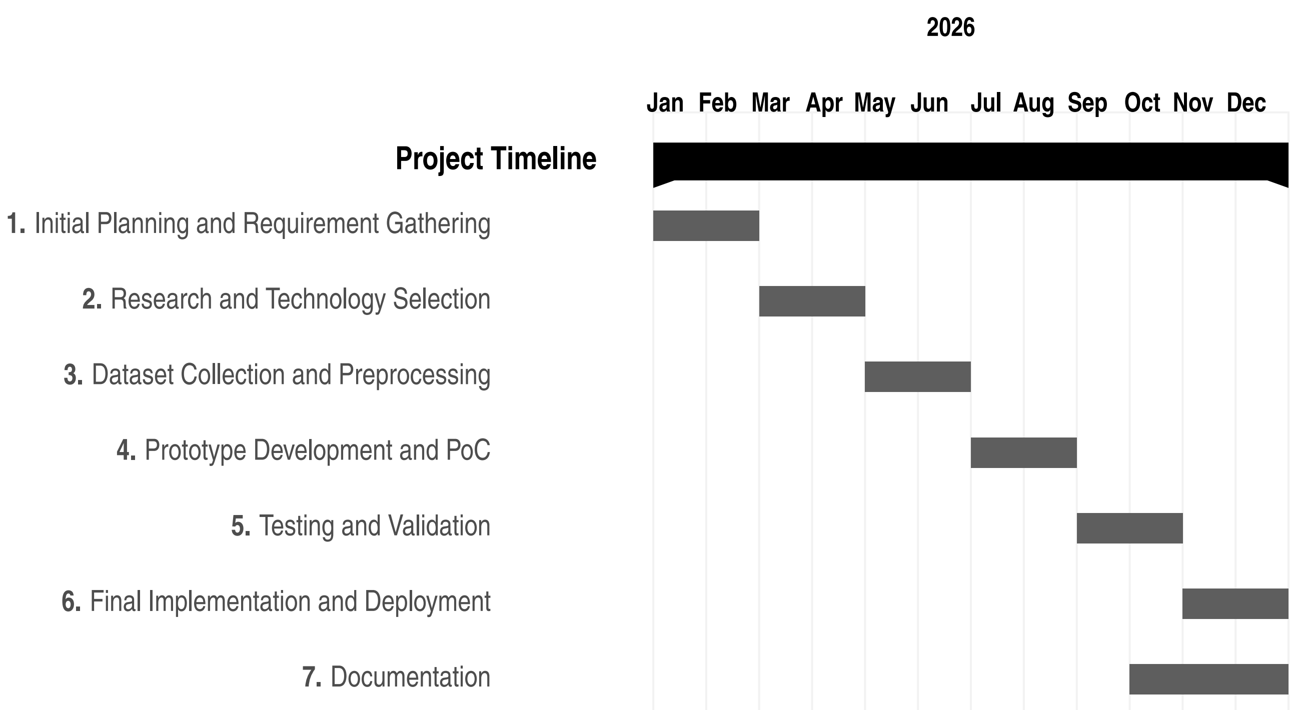

Table 4.1: Project Timeline

4.2 Team Organisation

The Cognix project is developed by a team of four members, each responsible for specific modules of the complete system.

4.2.1 Team Structure

1. **Komal — Module 1: Uncertainty Detection Engine**

   Komal is responsible for designing and implementing the Bayesian Neural Network-based uncertainty decomposition pipeline that forms the foundation of the entire Cognix system.

   Key Responsibilities:
   - ⋄ Aleatoric and Epistemic Uncertainty Decomposition using BNN
   - ⋄ Variational Inference training with ELBO loss
   - ⋄ Posterior sampling pipeline (S=50 weight samples)
   - ⋄ Structured output tuple generation (μ, σ_a, σ_e)
   - ⋄ CARLA simulation scenario design for validation
   - ⋄ DuckieBot hardware latency testing and optimisation
   - ⋄ Literature review for uncertainty detection methods

2. **Aryan — Module 2: Bayesian Belief Network**

   Aryan is responsible for multi-agent probabilistic belief fusion using Loopy Belief Propagation, weighted by epistemic uncertainty from Module 1.

   Key Responsibilities:
   - ⋄ Bayesian Belief Network design and implementation
   - ⋄ Loopy Belief Propagation algorithm
   - ⋄ Epistemic-uncertainty-weighted message passing
   - ⋄ Convergence testing and collective posterior computation

3. **Aditya — Module 3: Conformal Calibration Engine**

   Aditya is responsible for providing statistical guarantees on prediction reliability through conformal calibration techniques.

   Key Responsibilities:
   - ⋄ Conformal Prediction implementation with coverage guarantees
   - ⋄ Calibration of uncertainty outputs from Module 1 and Module 2
   - ⋄ Two-stage hierarchical calibration (individual and collective)
   - ⋄ Non-conformity score computation and threshold selection

4. **Zaid — Module 4 and 5: Communication and Decision Engine**

   Zaid is responsible for efficient inter-agent uncertainty communication and final risk-aware decision making with attribution.

   Key Responsibilities:

   - Graph Attention Network for uncertainty-aware communication
   - Bandwidth-constrained message selection protocol
   - Task-specific decision threshold design
   - Shapley value-based uncertainty attribution per agent
   - Escalation decision tree implementation
   - Final decision output formatting and logging

Chapter 5

Experimental setup

In order to evaluate the performance of the COGNIX system, a controlled experimental setup was designed. The aim was to analyze not only prediction accuracy but also the reliability and calibration of outputs. The system was tested under different conditions to understand how well it handles uncertainty in a multi-agent environment. The following sections describe the dataset, technologies used, and performance parameters considered during evaluation.

5.1 Data Set

Synthetic and Benchmark Dataset

- **Source:** Combination of synthetically generated data and standard machine learning benchmark datasets.
- **Formats:** Structured numerical data along with simulated observations representing different agent inputs.
- **Content Variety:**
  - ◇ Regression-based data for continuous prediction tasks
  - ◇ Classification scenarios with probabilistic outputs
  - ◇ Simulated multi-agent observations with varying uncertainty levels

5.2 Technology Used

- **Machine Learning Framework:**
  - ◇ PyTorch for building and training predictive models
- **Uncertainty Estimation:**
  - ◇ Monte Carlo Dropout and ensemble-based techniques for estimating uncertainty
- **Calibration Layer:**

- ◇ Conformal Prediction implemented as a post-processing step for generating calibrated intervals

- **Backend & Processing:**
  - ◇ Python 3.x for overall system implementation
  - ◇ NumPy and Pandas for data manipulation and processing
- **Development Tools:**
  - ◇ Git for version control and collaboration
  - ◇ Jupyter Notebook for experimentation and analysis

5.3 Performance Parameters

To evaluate the effectiveness of the system, both prediction performance and calibration quality were measured. The following parameters were considered:

- **Prediction Accuracy:**
  - Measures how close the predicted values are to actual values
  - Used standard metrics such as Mean Squared Error (MSE) or accuracy for classification
- **Calibration Accuracy:**
  - Evaluates how well the predicted confidence matches actual correctness
  - Goal: Confidence levels should align with real-world outcomes
- **Coverage:**
  - Percentage of true values that lie within the predicted confidence interval
  - Target: Coverage close to predefined confidence level (e.g., 95%)
- **Prediction Interval Width:**
  - Measures the size of the confidence interval
  - Narrower intervals with correct coverage indicate better performance
- **Computation Time:**
  - Time required to generate predictions and calibration intervals
  - Evaluated to ensure system efficiency
- **System Reliability:**
  - Measures consistency of predictions across different agents
  - Helps in evaluating robustness of multi-agent decision-making

Chapter 6

PO & PSO Mapping

6.1 Program Outcome (PO) Mapping

PO	Description	Level	Justification
PO1	**Engineering Knowledge:** Apply knowledge of mathematics, science, and engineering fundamentals to AI and DS problems.	3	Cognix applies advanced mathematics including Bayesian inference, probability distributions, Variational Inference, ELBO loss, Shapley values, graph theory, and conformal prediction to solve complex multi-agent uncertainty quantification problems.
PO2	**Problem Analysis:** Identify, formulate, and analyze complex problems using principles of math, natural science, and engineering.	3	The core problem of overconfident AI systems was identified, formally defined, and analyzed through a review of three recent IEEE papers, reaching the substantiated conclusion that aleatoric and epistemic uncertainty must be separated in real-time multi-agent settings.

PO3	**Design/Development of Solutions:** Design system components or processes considering societal, safety, and environmental factors.	3	A complete five-module system was designed addressing public safety — autonomous driving, medical diagnosis, financial trading — with appropriate consideration for regulatory compliance including ISO 26262 and EU AI Act 2.0.
PO4	**Conduct Investigations of Complex Problems:** Conduct research-based investigations, design experiments, analyze data, and synthesize conclusions.	3	Experiments are designed using CARLA simulator across four uncertainty scenario categories, with analysis of decomposition accuracy, ECE, latency, and OOD detection — all grounded in research-based investigation methodology.
PO5	**Modern Tool Usage:** Create and apply modern tools, modeling, and engineering techniques.	3	PyTorch 2.0.1, Pyro, BayesianTorch, CARLA 0.9.14, PettingZoo, DuckieBot, PyTorch Geometric, TorchScript, and Weights and Biases are all applied extensively across the Cognix system.
PO6	**The Engineer and Society:** Apply contextual knowledge to societal, health, safety, legal, and cultural issues.	3	Cognix directly addresses societal safety — autonomous vehicle accidents, medical AI misdiagnosis, financial flash crashes — and aligns with legal regulatory requirements of ISO 26262, EU AI Act 2.0, and FDA 2026 guidelines.

PO7	**Ethics:** Apply ethical principles and commit to professional ethics and responsibilities.	3	Cognix is fundamentally motivated by the ethical responsibility of deploying AI in safety-critical domains. The system ensures AI is transparent and honest about its confidence — not overconfident — before making decisions affecting human lives.
PO8	**Individual and Team Work:** Function effectively as an individual and in teams.	3	Cognix is developed by a four-member team with each member owning one module — functioning effectively both individually and collaboratively in a multidisciplinary AI and robotics project.
PO9	**Communication Skills:** Communicate effectively on complex engineering activities.	3	Technical documentation, IEEE literature review, seminar report, and PPT presentation were all prepared, communicating complex uncertainty quantification concepts clearly to both technical and non-technical audiences.
PO10	**Project Management and Finance:** Apply project management and financial principles in engineering work.	3	The project was structured across eight phases with clear module-wise division of work, timelines, integration milestones, and hardware procurement planning for DuckieBot and CARLA infrastructure.

PO11	**Life-long Learning:** Recognize the need for life-long learning amidst technological change.	3	The team independently studied cutting-edge research from 2022 to 2025 on Bayesian Neural Networks, conformal prediction, and graph-based uncertainty communication — all topics beyond standard curriculum — demonstrating self-directed life-long learning.

Table 6.1: Program Outcomes (PO) Mapping – Cognix

6.2 Program Specific Outcome (PSO) Mapping

PSO Number	**Description**	**Mapping Level**	**Justification**
PSO1	Interpret data and apply Artificial Intelligence algorithms to provide solutions.	3	Cognix interprets multi-modal sensor data from camera, LIDAR, and radar agents and applies Bayesian Neural Networks, Graph Neural Networks, Loopy Belief Propagation, and conformal prediction algorithms to provide safe and calibrated solutions for multi-agent decision making in autonomous driving, medical diagnosis, and distributed robotics.

PSO2	Apply standard practices, strategies and use appropriate models of data analytics to discover knowledge.	3	Standard Bayesian inference practices, Shapley value attribution analysis, and Expected Calibration Error evaluation strategies are applied to multi-agent sensor data to discover actionable knowledge — identifying which agents are reliable, which uncertainty sources can be reduced, and which scenarios require human intervention.

Table 6.2: Program Specific Outcomes (PSO) Mapping – Cognix

Chapter 7

Summary

Cognix is designed to solve a problem that existing AI systems largely ignore — the inability to explain why they are uncertain and what should be done about it. The system operates across five interconnected modules, where each agent first decomposes its sensor observations into two distinct uncertainty components, fuses its beliefs with neighbouring agents through probabilistic message passing, receives a mathematically guaranteed calibration ensuring stated confidence levels are accurate, communicates only the most informative uncertainty signals to other agents through a learned graph-based protocol, and finally receives a complete decision output that not only recommends an action but also identifies which specific agent is the weakest link in the collective reasoning chain. Applied to autonomous driving, medical diagnosis, and multi-robot coordination, Cognix moves AI decision making away from blind confidence toward a model of structured and explainable self-awareness, where every prediction carries a justified level of trust rather than an arbitrary score.

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